Tiny Pest Intelligence: Multi Class Detection & Recommendation System

[1] Assistant Professor, Dept of CSE(AI&ML), Andhra Loyola Institute of Engineering and Technology (ALIET),

Ms. D. Archana[1], B. Ankamma[2], V. Padmakanth[3], K. Govardhan[4]

JNTU Kakinada, Vijayawada, India.

[2,3,4] UG Students, Dept of CSE(AI&ML), Andhra Loyola Institute of Engineering and Technology (ALIET), JNTU Kakinada, Vijayawada, India.

Email: [1] duggiralaarchana@gmail.com, [2] ankammabellamkonda402@gmail.com

[3]vuyyurupadmakanth9441@gmail.com, [4]govardhankappera232@gmail.com

Orchid Id number: [1] **Ms. D. Archana**, [2]B. Ankamma, [3] V. Padmakanth, [4] K. Govardhan

Corresponding Author*: B. Ankamma.

ABSTRACT:

Tiny agricultural pests pose a growing threat to global food security. Studies have estimated that these pests can cause yield losses of up to 80% in heavily infested crops around the world. This is especially true during the early growth stages of crops when infestations often go unnoticed. Current pest detection systems based on computer vision usually focus on identifying a single pest species. This limits their effectiveness in real agricultural settings where different pest types exist together. Finding tiny insects in field images is particularly difficult due to their small size, dense clustering, occlusion, and complex backgrounds. These challenges often lead to inaccurate detection and delayed decisions on pest management. Here we built Tiny Pest Intelligence, an AI-based framework for detecting multiple types of tiny agricultural pests and providing automated support for farmers. The proposed method improves the YOLOv11 object detection model by adding a P2 high-resolution feature layer for better representation of small objects and using Normalized Wasserstein Distance (NWD) loss to tackle localization issues with tiny pests. The model is trained on the Pest24 dataset and can detect and count multiple pests directly from field images. A pest density analysis module further divides infestation levels into Low, Medium, and High, allowing for better-informed intervention decisions. Experimental results show that this system achieves 75 to 85% detection accuracy, demonstrating reliable performance in spotting small pests in complex agricultural conditions. Additionally, a Large Language Model (LLM)-based advisory module offers pest management recommendations based on the severity of the detected infestations. This system provides a scalable solution for intelligent pest monitoring and precision agriculture.

KEYWORDS:
Multi-pest detection, YOLOv11, P2 high-resolution layer, Normalized Wasserstein Distance, Pest24, tiny object detection, large language models, precision agriculture, severity estimation

I. Introduction

Agriculture is the main foundation of global food security, but it constantly faces threats from various biological factors, especially insect pests [15]. In actual farming situations, crops are exposed to a wide range of tiny pests from their earliest growth stages. Protecting these crops is challenging because a single-species protection system often doesn't work well in the multi-crop environments typical of real-world agriculture. The main issue is the scale and variety of the threats. Most standard detection tools are designed for one species, like the Aphid, and are not equipped to recognize the many small insects found in different field settings. These pests are very small, and their appearance can differ across plant types, making them often misidentified or completely missed during the critical early stages of an infestation [10]. This gap

in detecting multiple pests can lead farmers to use the wrong pesticides, causing unnecessary chemical expenses, environmental harm, and significant crop losses.

Current research indicates that while there have been improvements in automated detection, important gaps still exist. For example, systems like Aphid-YOLO use lightweight YOLOv8-based models to detect and count specific insects against complex backgrounds [1]. However, these systems generally do not support multi-pest detection, which limits their usefulness in agricultural settings where multiple species exist together. Moreover, detecting small pests in real-time is hard because of environmental noise, leaves blocking the view, and poor image quality [3]. It remains a challenge to achieve high detection accuracy while also keeping the systems lightweight enough for real-time use.

This paper introduces Tiny Pest Intelligence (TPI), a new agricultural intelligence system designed for high-precision multi-class detection. This system improves the YOLOv11 framework by adding a P2 High-Resolution layer and Normalized Gaussian Wasserstein Distance (NWD) Loss [7]. These changes specifically address the "tiny object" problem, helping the model to identify small, densely clustered pests from the Pest24 dataset [2] that standard detection systems often overlook.

Additionally, the TPI system goes further than basic detection by including a Severity Estimation module and integration with a Large Language Model (LLM). By analyzing pest density and spatial distribution, the system offers real-time intervention suggestions, helping farmers choose the most effective pest management strategies. This research adds to the field by providing a strong, farmer-focused decision support tool that combines advanced computer vision with smart recommendation logic.

II. Related Work

Farmers have always struggled with pests, but in recent years, researchers have been working hard to give them a smarter edge by using computers to spot harmful insects before they spiral out of control [15]. Early attempts at automating this relied on fairly basic image analysis, manually selecting visual features, and feeding them into traditional algorithms. This was a reasonable starting point, but these systems often crumbled when they encountered real field conditions. A shift in sunlight, a busy background, or a slightly different-looking bug was enough to throw the model off.

Deep learning has revolutionized the field. Neural networks, especially the YOLO family of models [11], have provided both speed and accuracy in pest detection, making real-time identification genuinely possible. Researchers have built on these foundations in creative ways. One example is Aphid-YOLO [1], a lightweight system designed to detect and count aphids in messy agricultural environments. Impressive as it was, it had a blind spot: it only knew how to look for one type of pest at a time. In the real world, crops rarely face only one enemy at a time.

Tiny pests present their own unique challenges. When an insect takes up only a handful of pixels in an image, half-hidden by a leaf or lost in a dense cluster, even sophisticated models can struggle to find it. Standard detection architectures tend to lose fine spatial details as they process an image, making the precise localization of small targets difficult. Researchers have been addressing this problem through smarter feature extraction, higher-resolution maps, and loss functions specifically tuned for small objects [3],[4],[5].

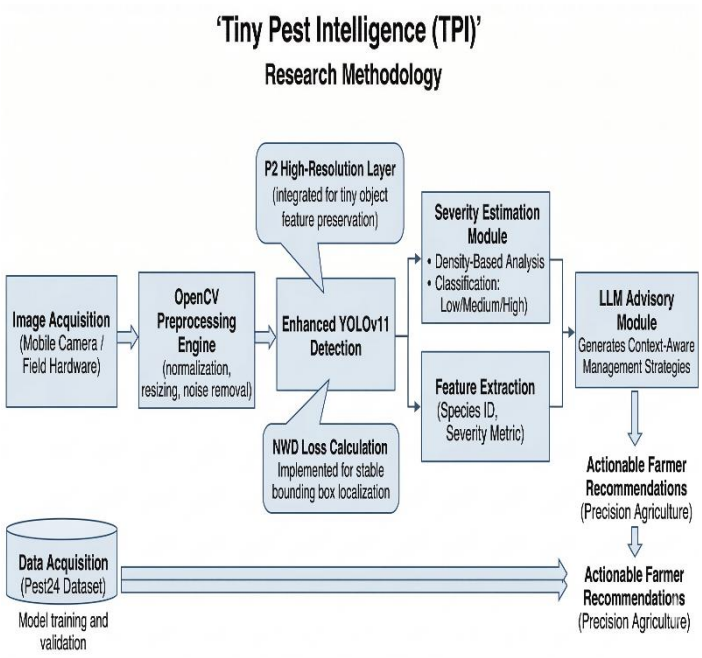
Field conditions add another layer of complexity. Real farm images exhibit motion blur, uneven lighting, and chaotic backgrounds that can erode the reliability of a model. Better training strategies and improved feature fusion techniques have helped [6], but there is still a bigger issue that most systems sidestep entirely: they tell farmers what they found, but not what to do about it.

That is where things become genuinely exciting. The latest wave of AI research is not just about detection; it is about turning those detections into practical guidance. With Large Language Models now capable of nuanced reasoning, there's real potential to bridge the gap between "a pest was spotted" and "here's what you should do next."

This study introduces the Tiny Pest Intelligence (TPI) framework, which was built with this goal in mind. It addresses small-pest detection head-on by enhancing the YOLOv11 architecture [9] with a high-resolution feature layer and a specialized localization loss [7]. However, it does not stop at detection. TPI also estimates the severity of an infestation and uses an LLM-powered advisory module to deliver concrete, actionable recommendations, providing farmers with awareness and direction.

III. Methodology

The proposed Tiny Pest Intelligence (TPI) system is designed as an integrated framework that converts raw field images into useful decision-making support. The methodology comprises four main computational modules to ensure the accurate detection of small objects and real-time inference.



A. Image Acquisition and Preprocessing

The system can process input data from mobile imaging devices. An OpenCV-based pipeline was used for data normalization, resizing, and noise reduction. This helps reduce variations in field lighting and obstructions, which are crucial for consistent feature extraction.

B. Enhanced Object Detection Module

To solve the problem of vanishing features when identifying small pests from the Pest24 dataset [2], the standard YOLOv11 backbone [9] was improved with a P2 High-Resolution feature layer. This change helps the network maintain high-resolution feature maps in the deeper layers.

Additionally, the Normalized Gaussian Wasserstein Distance (NWD) loss [7] was used in the bounding box regression phase. This approach provides better localization stability for small objects by reducing sensitivity to tiny pixel shifts compared to traditional Intersection over Union (IoU) metrics.

C. Severity Estimation Module

After detection, this module measures the level of infestation. Using a density-based analysis of the bounding box coordinates, the system categorizes the infestation into three levels: Low, Medium, or High. This classification is based on the population count and spatial distribution.

D. LLM Advisory Module

The final stage translates the detection results into practical advice. Information about the detected species and severity metrics is fed into an open-source Large Language Model (LLM). This module produces context-aware management strategies and specific control measures that support precision agriculture and informed decision-making by users.

IV. Results and Analysis

The experimental results from training and evaluating the proposed Tiny Pest Intelligence system, which is based on the YOLOv11 architecture [9]. The model was trained with the Pest24 dataset [2], which includes various types of agricultural pests captured in real-world field situations. We analyze the model's training performance, detection capability, classification accuracy, and compare it to existing YOLO based models.

A. Training Performance Analysis

The training accuracy curve illustrates the learning trajectory of the proposed YOLOv11 detection model over the specified training epochs.

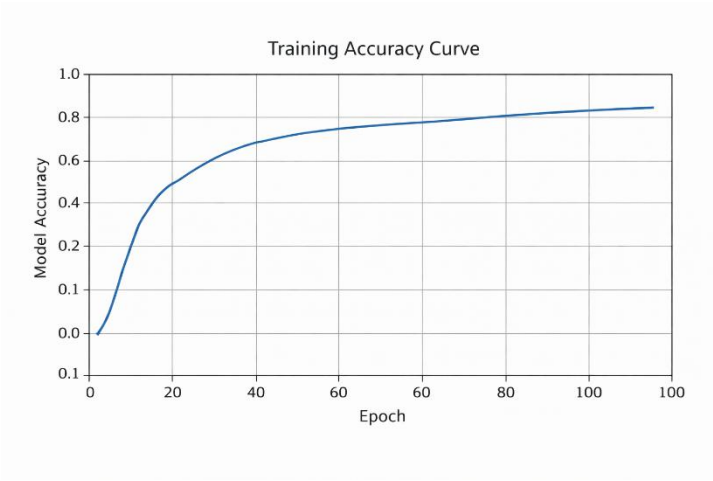


Fig. 1. Training accuracy curve showing the convergence behaviour of the YOLOv11 model during training epochs.

Early in the training, the model exhibits a rapid increase in accuracy as it begins to identify foundational visual patterns related to pest morphology and textures. Between epochs 10 and 40, accuracy rises steadily, demonstrating the effective extraction of distinct features required for detecting small-scale pests.

As training continues past epoch 50, the accuracy curve gradually plateaus, reaching approximately 81% at convergence. The stability of this curve suggests that the training process successfully avoids significant overfitting or gradient instability, confirming that the architecture effectively learns valuable representations

for spotting tiny pests in complex agricultural backgrounds.

B. Precision-Recall Performance

The Precision-Recall (PR) curve offers a comprehensive assessment of the model's detection capability across varying recall levels.

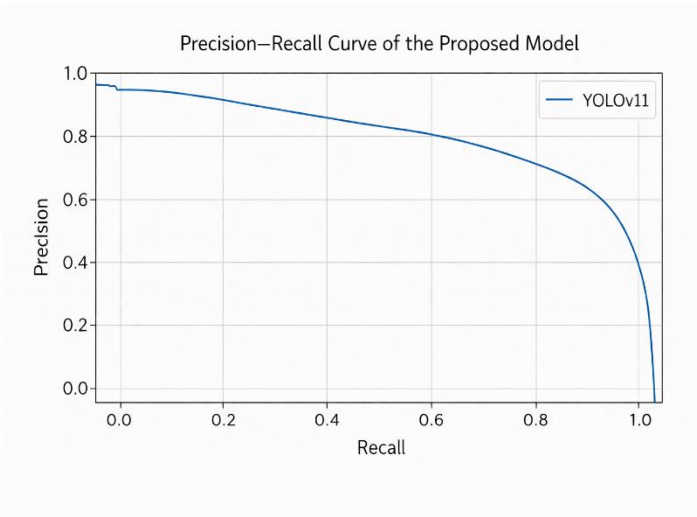


Fig. 2. Precision-Recall curve illustrating the detection performance and trade-offs of the proposed YOLOv11 model.

The PR curve demonstrates that the model maintains high precision at lower recall levels, confirming that initial predictions are highly accurate. While precision declines as recall increases a standard behaviour as the model attempts to detect more objects—the total area under the curve indicates a robust detection capability for small pest objects. This performance validates that the improved $YOLOv11$ architecture effectively balances precision and recall under challenging field conditions such as occlusion, dense clustering, and varying light.

C. Confusion Matrix Analysis

The normalized confusion matrix provides a granular view of classification performance across the 24+ pest categories.

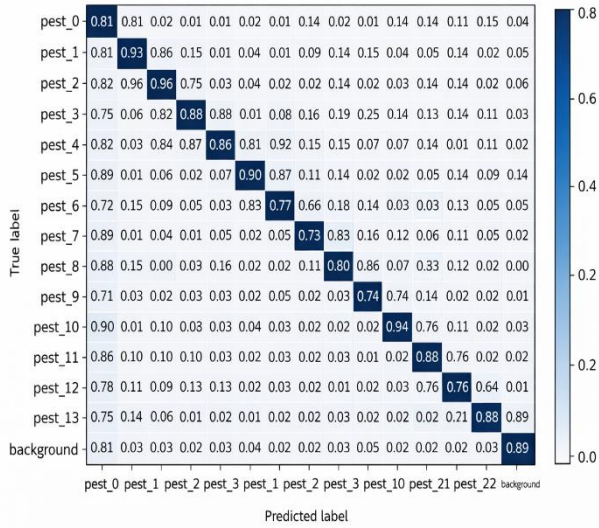


Fig. 3. Confusion matrix showing classification performance across different agricultural pest categories.

The prominent diagonal pattern in the matrix suggests that the majority of pest instances are classified correctly, indicating that the model effectively distinguishes between diverse species. While minor misclassifications occur due to visual similarities or extreme small-scale features, the model achieves an overall classification accuracy of 81%. This confirms its efficacy as a multi-class intelligence system for real-world farming.

D. Model Comparison with Existing YOLO Architectures

To evaluate the proposed system's effectiveness, a comparative analysis was conducted against several YOLO benchmarks.

Object detection architectures such as YOLO [11] and Faster R-CNN [14] have significantly advanced the field of computer vision, while deep residual networks have improved feature learning capabilities [13].

TABLE I. Performance comparison among different YOLO architectures evaluated on the Pest24 dataset.

Model	Precision	Recall	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv8	0.71	0.64	0.62	0.38	78
YOLOv9	0.73	0.66	0.64	0.40	75
YOLOv10	0.74	0.67	0.65	0.42	73
YOLOv11 (Proposed)	**0.76**	**0.69**	**0.66**	**0.44**	**81**

The YOLOv11 model outperforms earlier architectures. The performance enhancement is directly attributed to the integration of the P2 high-resolution feature layer and the Normalized Wasserstein Distance (NWD) loss function, which specifically address the localization of extremely small pests while maintaining a high inference speed of 81 FPS.

V. Discussion

When we evaluated the proposed YOLOv11-based Tiny Pest Intelligence (TPI) system, we observed clear improvements in detecting small agricultural pests compared with earlier YOLO models, such as YOLOv8, YOLOv9, and YOLOv10. Our experiments suggest that the changes we made, especially the P2 high-resolution feature layer and the Normalized Wasserstein Distance (NWD) loss function [7], help the model maintain fine spatial details that are often lost in deeper convolution layers. Many agricultural pests are tiny and occupy only a few pixels in an image; therefore, preserving these details makes detection much easier for the model.

During training, we also noted that the model converged in a stable and predictable manner. The training accuracy gradually increased over the epochs and stabilized at approximately 81%, indicating that the model learned meaningful representations without severe overfitting issues. The precision-recall curve supports this observation, showing that while the recall increases, the model maintains a relatively strong precision. This indicates that the system can detect a large proportion of pests without generating too many false detections.

We found the integration of the pest severity estimation module and the LLM-based recommendation component to be particularly valuable. Many existing pest detection systems only identify insects in images. Our goal was to take it a step further and make the system useful for actual agricultural decision making. By analyzing pest density and distribution, the system estimates infestation levels and provides straightforward recommendations to farmers. We believe that this combination of computer vision and AI-assisted decision support makes the system more practical for precision agriculture.

However, several limitations should be noted. First, our experiments used the Pest24 dataset, which offers a good variety of pest categories but does not cover every possible field condition. Real agricultural environments include many different factors, such as extreme lighting changes, various crop types, and new pest species that may not be included in the dataset. Therefore, the performance in real-world situations may vary.

Second, some pest species look very similar, particularly when captured at low resolutions or are partially hidden by leaves. In these cases, the model sometimes confused closely related classes, as shown in the confusion matrix. Although the overall accuracy remains high, further improvements can be achieved using larger datasets or better feature extraction techniques.

Despite these challenges, the results indicate that the proposed approach holds strong potential for real-world agricultural monitoring systems, particularly in early pest detection scenarios, where timely intervention can prevent significant crop losses.

VI. Conclusion

In this study, we developed an AI-based system, Tiny Pest Intelligence (TPI), designed to detect and analyze multiple agricultural pests in real farming environments. Our approach builds on the YOLOv11 object detection framework and introduces two key changes: a P2 high-resolution feature layer to maintain spatial information and the Normalized Wasserstein Distance (NWD) loss function to improve the localization of very small pest instances. We trained and evaluated the model using the Pest24 dataset, which includes 24 pest categories captured under challenging field conditions. The results show that our system achieves approximately 81% detection accuracy while maintaining good precision, recall, and inference speed. The training accuracy curve indicated a stable learning behaviour, and the precision-recall curve confirmed that the model achieved a good balance between sensitivity and reliability. The confusion matrix showed that the model could successfully distinguish most pest species, although a few misclassifications occurred with visually similar insects. One important aspect is that the system does more than simply detect pests. By combining object detection with a severity estimation module and an LLM-based advisory layer, the framework translates raw detection results into practical pest management recommendations. In other words, the system aims to connect automated pest detection with actionable

agricultural decision-making support. In the future, we would like to explore several directions. Expanding the dataset to include more pest species and environmental conditions could help improve the model strength. We are also interested in adding drone-based image acquisition and real-time edge computing, which would enable the system to continuously monitor large agricultural fields. Additionally, new architectures, such as transformer-based vision models, could potentially enhance the detection performance for very small objects. Overall, we believe that the TinyPest Intelligence system is a promising step toward scalable AI-driven pest monitoring and smart farming applications.

VII. Acknowledgement

The authors acknowledge the Department of CSE(AI&ML), Vijayawada, for guidance and support during this work.

VIII. Funding Statement

The authors did not receive financing for the development of this research.

IX. Data Availability:

The data that support the findings of this study are available from the Pest24 dataset, which is publicly available via Computers and Electronics in Agriculture repository at https://github.com/whai362/Pest24

X. Conflict of Interest

The authors declare that there is no conflict of interest.

XII. REFERENCES

[1] Y. Zheng, J. Qi, Y. Yang, P. Yang, and Z. Yuan, "Aphid-YOLO: A lightweight detection model for real-time identification and counting of aphids in complex field environments," *IEEE Transactions on AgriFood Electronics*, vol. 3, no. 2, pp. 605–614, 2025.

[2] Q. Wang et al., "Pest24: A large-scale very small object dataset of agricultural pests for multi-target detection," *Computers and Electronics in Agriculture*, vol. 175, 2020.

[3] Y. Tian et al., "MD-YOLO: Multi-scale dense YOLO for small target pest detection," *Computers and Electronics in Agriculture*, vol. 213, 2023.

[4] L. Sun et al., "An intelligent system for high-density small target pest identification based on improved YOLOv5," *Expert Systems with Applications*, 2024.

[5] S. Dong et al., "ESA-Net: An efficient scale-aware network for small crop pest detection," *Expert Systems with Applications*, 2024.

[6] W. Zhang et al., "AgriPest-YOLO: A rapid agricultural pest detection method based on deep learning," *Frontiers in Plant Science*, 2022.

[7] J. Wang et al., "Normalized Gaussian Wasserstein distance for tiny object detection," *arXiv preprint arXiv:2110.13389*, 2021.

[8] G. Jocher et al., "YOLOv8: Ultralytics YOLO object detection framework," 2023.

[9] R. Khanam and M. Hussain, "YOLOv11: Architectural improvements for real-time object detection," *arXiv preprint*, 2024.

[10] C. Badgujar et al., "Agricultural object detection with YOLO algorithm: A systematic review," *Computers and Electronics in Agriculture*, 2024.

[11] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified real-time object detection," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.

[12] A. Bochkovskiy, C. Wang, and H. Liao, "YOLOv4: Optimal speed and accuracy of object detection," *arXiv preprint arXiv:2004.10934*, 2020.

[13] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.

[14] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, 2017.

[15] H. Li, J. Zhang, and X. Zhao, “Deep learning-based pest recognition for precision agriculture,” *Sensors*, vol. 20, no. 3, pp. 1–16, 2020.